OCR Further Mathematics
Pure Core - Matrices
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ kx \end{pmatrix} = \begin{pmatrix} 2x+kx \\ -x \end{pmatrix}$ same line ⇒ $-x = k(2x + kx)$ for all x ($\neq 0$) ⇒ $-1 = k(2 + k)$ ⇒ $k^2 + 2k + 1 = 0$ ⇒ $k = -1$ (i.e. $y = -x$)	M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 1.1) [4]	Value of k can be implied by the correct equation	
	b		$\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ -x \end{pmatrix} = \begin{pmatrix} 2x-x \\ -x \end{pmatrix} = \begin{pmatrix} x \\ -x \end{pmatrix}$ so each point maps to itself and it is a line of invariant points	B1 (AO 2.4) [1]	Must have a reason e.g. it is sufficient to test one point other than (0, 0)	
			Total	5		
2	a		$\mathbf{A}^2 = \begin{pmatrix} 4 & 12 \\ 0 & 4 \end{pmatrix}$, $\mathbf{A}^3 = \begin{pmatrix} 8 & 36 \\ 0 & 8 \end{pmatrix}$, $\mathbf{A}^4 = \begin{pmatrix} 16 & 96 \\ 0 & 16 \end{pmatrix}$ Conjecture: $\mathbf{A}^n = \begin{pmatrix} 2^n & 3n \times 2^{n-1} \\ 0 & 2^n \end{pmatrix}$	B1 (AO2.2a) B1 (AO2.2b) [2]	BC Allow this mark for any conjecture which works for n = 1, 2, 3 and 4.	
	b		Basis case: n = 1:	B1 (AO2.1) M1 (AO2.1) M1	Allow this mark even if the conjecture is wrong, provided that it	A formal proof by induction is required for full marks.

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\mathbf{A}^1 = \begin{pmatrix} 2^1 & 3\times1\times2^0 \\ 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix} = \mathbf{A}$ so true for n = 1	(AO2.2a)	works for n = 1	
			Assume true for n= k ie $\mathbf{A}^k = \begin{pmatrix} 2^k & 3k\times2^{k-1} \\ 0 & 2^k \end{pmatrix}$	A1 (AO2.4)	Must have statement in terms of some other variable than n. Conjecture need not be correct.	
			$\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \begin{pmatrix} 2^k & 3k\times2^{k-1} \\ 0 & 2^k \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$ $= \begin{pmatrix} 2^{k+1} & 3\times2^k + 3k\times2^k \\ 0 & 2^{k+1} \end{pmatrix}$ $= \begin{pmatrix} 2^{k+1} & 3(k+1)\times2^k \\ 0 & 2^{k+1} \end{pmatrix}$	[4]	Uses inductive hypothesis properly & expands	
			So true for n = k⇒ true for n = k + 1. But true for n = 1. So true for all positive integer n		AG. Manipulating terms correctly and convincingly to obtain required form. Some intermediate working must be seen and a clear conclusion must be given for the induction process.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	6	

Question			Answer/Indicative content	Marks	Part marks and guidance		
3	a		DR A shear which leaves the x-axis invariant and which transforms the point (0, 1) to the point (2, 1).	B1 (AO2.2a) [1]	Or any useful point transformed to its image	not "scale factor" or sf	
	b		DR detA = 1×1 – 0×2 = 1 and this is the area scale factor	B1 (AO2.4) [1]	Both	Detailed calculation must be shown	
	c		DR $\begin{pmatrix}1&0\\3&1\end{pmatrix}$ seen $\begin{pmatrix}1&2\\0&1\end{pmatrix}\begin{pmatrix}1&0\\3&1\end{pmatrix}=\begin{pmatrix}7&2\\3&1\end{pmatrix}$ $\begin{pmatrix}1&0\\0&p\end{pmatrix}\begin{pmatrix}7&2\\3&1\end{pmatrix}$ $=\begin{pmatrix}7&2\\3p&p\end{pmatrix}\Rightarrow p=7$	B1 (AO3.1a) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	BC Correct form for stretch multiplied into their matrix in either order Correct multiplication		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
4			$\begin{pmatrix} -12 & -1 & 5 \\ -1 & -20 & 3 \\ 7-6a & 3a-2 & -4a-5 \end{pmatrix}$ or $\begin{pmatrix} -4a & -1 & 5 \\ 11-4a & -20 & 3 \\ -8-a & 7 & -17 \end{pmatrix}$ seen	M1 (AO1.1)	Either product AB or BA calculated (but not if assigned incorrectly). Alternatively: equivalent correct useful entries calculated for both	Condone 3 errors or omissions
			$-12 = -4a$ or $-1 = 11 - 4a$ or $7 - 6a = -8 - a$ or $3a - 2 = 7$ or $-4a - 5 = -17$	M1 (AO1.1)		This mark can be implied by sight of a correct equation
			$a = 3$	A1 (AO2.2a)	Finding matrix products both ways and equating entries usefully	This mark can be implied by sight of a correct equation even if other entries or equations are wrong. Cannot be awarded if either AB or BA has more than 3 errors
				[3]		
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
5	a		$\mathbf{A}=\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix}$ $\mathbf{A}=\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix}\begin{pmatrix}2\\ 1\end{pmatrix}=\begin{pmatrix}2\\ 2k+1\end{pmatrix}=\begin{pmatrix}2\\ 9\end{pmatrix}$ $\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix}\begin{pmatrix}2\\ 1\end{pmatrix}=\begin{pmatrix}2\\ 2k+1\end{pmatrix}=\begin{pmatrix}2\\ 9\end{pmatrix}$ $k = 4$ so $\mathbf{A}=\begin{pmatrix}1 & 0\\ 4 & 1\end{pmatrix}$	B1(AO1.2) M1(AO1.1) A1(AO1.1)[3]	Correct form of A seen Correctly multiplying object vector into their A to find image and equating to given image <u>Examiner's Comments</u> Most candidates successfully found A. Many benefited from the ability to recall the general form of the matrix representing a shear, y-axis invariant. They used this to find the unknown element by correctly multiplying their matrix by the object vector and equating to the image vector. Candidates who had not learnt the shear matrix forms often started from a more general 2x2 matrix with two or four unknown elements (depending on their conceptual grasp of invariance). Frequently those trying to determine four unknowns were unsuccessful. Other errors sometimes observed were attempts to right-multiply by A or consideration of the stretch matrix with zeros in the leading diagonal.	their A must have at least 1 unknown element	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		detA = 1 − 0 = 1 The determinant is the area scale factor so a determinant of 1 leaves the area unchanged	B1(AO1.1)B1(AO2.4)[2]	Correctly finding determinant Convincing explanation. Must include both ideas. <u>Examiner's Comments</u> For candidates that had answered part (a), finding the determinant generally presented no problems. To gain the second mark here, candidates needed to explain that the (absolute value of) the determinant gives the area scale factor of the transformation. Without the word 'area', the candidate could be mistakenly talking about a length scale factor (which, of course, also yields no change in area).	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6	a		$\det A$ (= 0.6 × 1.8 – –0.8 × 2.4) = 3	B1 (AO1.1) [1]		
	b		Determinant of rotation = 1 Determinant of rotation × determinant of stretch = 1 × sf = 3 => sf = 3	B1 (AO1.1) B1 (AO2.2a) [2]		
	c		Since the second column of A contains entries bigger than 1 (in magnitude) the stretch must be parallel to the y-axis.	B1 (AO2.4) [1]	Or any correct, complete explanation.	May see $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 3\end{pmatrix}$ $=\begin{pmatrix}\cos\theta & -3\sin\theta\\ \sin\theta & 3\cos\theta\end{pmatrix}$ or similar
	d		$\sin\theta = -0.8$ and $\cos\theta = 0.6$ oe awrt −53° (or −0.93 rads)	M1 (AO2.2a) A1 (AO1.1) [2]	Condone if only one equation or 53° (0.93 rads) clockwise or 307° (5.36 rads) (anticlockwise).	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7	a		$A^4 = I$	B1 (AO1.1) [1]	Accept 3 × 3 matrix Examiner's Comments In part (a), candidates should have noted the command word 'Find' alongside the single mark allocation which signifies that no working need be seen and that their calculator should be used. Those that squared the matrix A and then squared the result may have lost time. OCR support Candidates should understand the requirements expected where a question uses command words such as 'Find', 'Solve' or 'Calculate'. These command words indicate, while working may be necessary to answer the question, no justification is required and a solution could be obtained from the efficient use of a calculator. The command words are defined in the specification and a student summary guide can be found on the assessment tab of the H245 qualification page on the OCR website. A Level Maths command words poster.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		Rotation Clockwise 90° about x-axis	B1 (AO2.2a) B1 (AO2.2a) [2]	Or 270° anticlockwise Accept radians Examiner's Comments In part (b) that the transformation represented a rotation earned one mark with the other mark being given for a full explanation of what type of rotation it was.	
	c		$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	B1 (AO1.1) [1]	All correct	
	d		(−2, 3, 4)	B1 (AO1.1) [1]	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$ Allow vector as answer	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8	a		$\det A = a^2 - 10a + 16$	M1(AO1.1a) A1(AO1.1) [2]	Attempt to work out the determinant <u>Examiner's Comments</u> These two parts were answered well, but a few thought that the value of the determinant was the reciprocal, i.e. $\frac{1}{a^2 - 10a + 16}$.	
	b		$a^2 - 10a + 16 = 0 \Rightarrow (a - 2)(a - 8) = 0 \Rightarrow a = 2, 8$	M1(AO1.1a) A1(AO1.1) [2]	Solving *their* quadratic soi <u>Examiner's Comments</u> For candidates that had obtained the right quadratic equation in (a), the two values were obtained very easily.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		For both values there is no unique solution as detA = 0 For a = 2, equations are: p_1: $2x + 2y = 6$ p_2: $2y + 2z = 8$ p_3: $4x + 5y + z = 16$ $2p_1 + \frac{1}{2}p_2 = p_3$ So there is an infinite set of solutions. For a = 8, equations are: p_1: $8x + 2y = 6$ p_2: $8y + 2z = 8$ p_3 : $4x + 5y + z = 16$ $\frac{1}{2}p_1 + \frac{1}{2}p_2 \neq p_3$ as it gives $4x + 5y + z = 7$ and $16 \neq 7$ so no solution	B1(AO2.4) M1(AO2.1) A1(AO1.1) A1(AO2.1a) M1(AO2.1) A1(AO1.1) A1(AO2.2a) [7]	Soi by correct answers Substitute one of *their* values and solve Substitute the other one of *their* values and solve <u>Examiner's Comments</u> Many candidates were able to access full marks for this part. However, in a significant number of scripts the logical reasoning leading to the answer was lacking. The command word 'determine' requires reasoning and justification; sometimes it was hard to navigate through candidate's working to discern its accuracy or otherwise.	"correct answers" means solns are either infinite or nonexistent.	
			Total	11			

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		a	$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$	B1 (AO 1.1) [1]		
		b	$\frac{1}{4}\begin{pmatrix} 1 & 0 & 1 \\ -8 & 4 & 0 \\ 19 & -8 & -1 \end{pmatrix}$ **oe**	B1 (AO 1.1) [1]		
		c	(i) E.g. Because expressed in matrix form the system of equations is AX = C where A is the matrix is A from above E.g. And it has been established that an inverse exists. (ii) $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 2 \\ 3 & -2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$ $\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 \\ -8 & 4 & 0 \\ 19 & -8 & -1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$ $\Rightarrow x = 1, y = 1, z = -3$	B1 (AO 2.4) B1 (AO 2.4) [2] M1 (AO 3.1a) A1 (AO 1.1) [2]	Use B and multiply May be seen in part (i) BC	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		a	$3x + 4y = 28$ so $a = 3$, $b = 4$, $c = 28$ (or any non-zero multiples) so $D = \frac{\lvert 3 \times -6 + 4 \times 4 - 28 \rvert}{\sqrt{3^2 + 4^2}}$	M1 (AO 1.1)	Identifying a, b and c and substituting a, b, c and (x_1, y_1) correctly into distance formula	
			$D = 6$	A1 (AO 1.1)		
			Alternative solution $y - 4 = \frac{4}{3}(x + 6)$ oe so $-0.75x + 7 - 4 = \frac{4}{3}(x+6) \Rightarrow x = -2.4,\ y = 8.8$	M1	Finding equation of perpendicular line through (–6, 4) and solving simultaneously to find foot of perpendicular	
			$D = \sqrt{(-2.4 - -6)^2 + (8.8 - 4)^2} = 6$	A1		
				[2]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$\begin{pmatrix}2\\1\\-4\end{pmatrix}\times\begin{pmatrix}3\\-1\\1\end{pmatrix}=\begin{pmatrix}-3\\-14\\-5\end{pmatrix}$ $D=\dfrac{\left\|\left(\begin{pmatrix}11\\-1\\5\end{pmatrix}-\begin{pmatrix}4\\3\\-2\end{pmatrix}\right).\begin{pmatrix}-3\\-14\\-5\end{pmatrix}\right\|}{\left\|\begin{pmatrix}-3\\-14\\-5\end{pmatrix}\right\|}$ or $\dfrac{\left\|\begin{pmatrix}7\\-4\\7\end{pmatrix}.\begin{pmatrix}-3\\-14\\-5\end{pmatrix}\right\|}{\left\|\begin{pmatrix}-3\\-14\\-5\end{pmatrix}\right\|}$ $D = 0$ Alternative solution $4 + 2\lambda = 11 + 3\mu$, $3 + \lambda = -1 - \mu$ and $-2 - 4\lambda = 5 + \mu$ $\lambda = -1, \mu = -3$ eg $-2 - 4(-1) = 2 = 5 + -3$ so lines intersect so $D = 0$	B1 (AO 1.1a) M1 (AO 1.1) A1 (AO 1.1) M1 A1 A1 [3]	Correctly finding a mutual perpendicular BC Correct substitution into distance formula Looking for a PoI so all 3 (3rd might be seen later) Correctly solving any 2 equations Must be checked in the unsolved equation.	Value of each side must be found, not just equality asserted.

Question			Answer/Indicative content	Marks	Part marks and guidance
		c	There are two points, one on each line, such that the distance between the points is 0… ...and so the lines must intersect.	E1ft (AO 3.1a) E1ft (AO 2.4) [2]	If *D* found to be non-zero in (b) then allow "Because there are not two points…" A convincing demonstration that the two direction vectors are not parallel and "...and so the lines must be skew"
			Total	7	

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		a	$a = 1$ $\frac{1}{-24}\begin{pmatrix} 24 & -12 & 0 \\ -48 & 15 & 6 \\ 16 & -6 & -4 \end{pmatrix}\begin{pmatrix} 6 \\ 8 \\ k \end{pmatrix}$ $x = -2,\ y = 7 - \frac{1}{4}k,$ $z = -2 + \frac{1}{6}k$	B1 (AO 2.2a) M1 (AO 1.1a) A1 (AO 1.1) [3]	soi	
		b	$a = 2$ $b = 5$ $4x - 6z = 18$ $z = \frac{2}{3}x - 3$ $y = 7 - 2x$	B1 (AO 2.2a) B1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 1.1) A1 (AO 1.1) [5]	soi Use of $2x + 2y + 3z = 5$ and $-2x + 2y + 9z = -13$ to eliminate y or z	
		c	The solution represents a straight line E.g. Two of the planes are identical and the third intersects it/them (in a straight line)	B1 (AO 2.2a) B1 (AO 3.2a) [2]		
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	$\mathbf{AB} = \begin{pmatrix} 1 & a \\ 3 & 0 \end{pmatrix}\begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 4+3a & 2+3a \\ 12 & 6 \end{pmatrix}$ $\mathbf{BA} = \begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & a \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} 10 & 4a \\ 12 & 3a \end{pmatrix}$ Equate any element give a = 2	M1 (AO2.1) A1 (AO1.1) A1 (AO1.1) [3]		
		b	Choose any two matrices that are not commutative	M1 (AO2.1) A1 (AO2.1) [2]	Choose Demonstrate BC	
		c	Det B = 6 gives area = 24	B1 (AO1.1) [1]		
		d	$\begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ m \end{pmatrix} = \begin{pmatrix} \lambda \\ \lambda m \end{pmatrix}$ $\Rightarrow 4 + 2m = \lambda,\ 3 + 3m = \lambda m$ $\Rightarrow 3 + 3m = (4 + 2m)m$ $\Rightarrow 2m^2 + m - 3 = 0 \Rightarrow (m - 1)(2m + 3) = 0$ $\Rightarrow m = 1$ and $m = -\frac{3}{2}$ i.e. $2y + 3x = 0$ and $y = x$	M1 (AO2.1) A1 (AO1.1) A1 (AO1.1) A1 (AO2.2a) [4]		
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
13		a	$\begin{pmatrix} 3 & 2 & -1 \\ 2 & -4 & 7 \\ 10 & 20 & -25 \end{pmatrix}^{-1} = \frac{1}{40}\begin{pmatrix} -40 & 30 & 10 \\ 120 & -65 & -23 \\ 80 & -40 & -16 \end{pmatrix}$ oe	M1 (AO 1.1a)	BC	eg $\begin{pmatrix} -1 & \frac{3}{4} & \frac{1}{4} \\ 3 & -\frac{13}{8} & -\frac{23}{40} \\ 2 & -1 & -\frac{2}{5} \end{pmatrix}$	
			$\frac{1}{40}\begin{pmatrix} -40 & 30 & 10 \\ 120 & -65 & -23 \\ 80 & -40 & -16 \end{pmatrix}\begin{pmatrix} 5 \\ 60 \\ b \end{pmatrix} = \begin{pmatrix} 40+\frac{1}{4}b \\ -\frac{3300+23b}{40} \\ -50-\frac{2b}{5} \end{pmatrix}$	A1 (AO 1.1)	For any correct component		
			$x = 40+\frac{1}{4}b,\ y = -\frac{3300+23b}{40},\ z = -50-\frac{2b}{5}$	A1 (AO 1.1) [3]	All three correct (oe)		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$\begin{vmatrix} 3 & 2 & -1 \\ 2 & -4 & 7 \\ a & 20 & -25 \end{vmatrix}$ $= 3(100 - 140) - 2(-50 - 7a) - 1(40 + 4a)$ $10a - 60$ $a = 6$	M1 (AO 3.1a) A1 (AO 1.1) A1 (AO 2.2a) [3]	Consideration of determinant of correct matrix and genuine attempt to find determinant. Set to 0 and solve	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	(i) $2\alpha - 4\beta = 20$ or $-\alpha + 7\beta = -25$	M1 (AO 2.2a)	or any valid method to find α and β	
			$\begin{pmatrix} 2 & -4 \\ -1 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 20 \\ -25 \end{pmatrix} = \frac{1}{10}\begin{pmatrix} 7 & 4 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 20 \\ -25 \end{pmatrix}$	M1 (AO 1.1)		
			$\alpha = 4, \beta = -3$	A1 (AO 1.1)		
				[3]		
			(ii) $4 \times 5 + (-3) \times = 60$	M1 (AO 2.2a)	Their α and β.	
			-160	A1 (AO 1.1)		
				[2]		
			(iii) The three planes form a sheaf with a common line of intersection. oe	E1 (AO 3.2a)	Do not ISW (eg if 'or two of the planes may be parallel' then E0).	
				[1]		
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 7 & 11 & 12 \end{pmatrix}^{-1} \begin{pmatrix} 3 \\ 9 \\ 20 \end{pmatrix}$ $x = 5, y = -9, z = 7$	M1(AO 1.2) A1(AO1.1) [2]	BC or in vector form	OR elimination method M1 reduce to two equations in two unknowns	
		b	$\mathbf{M} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 7 & 11 & p \end{pmatrix}$ Det M = $4p + 35 + 22 - 28 - 2p - 55$ M is singular, so $2p - 26 = 0$ $\Rightarrow p = 13$ The third equation must be a linear combination of the first two $\alpha + 2\beta = 7$, $\alpha + 4\beta = 11$, $\alpha + 5\beta = 13$ $\Rightarrow \alpha = 3$, $\beta = 2$ $3\alpha + 9\beta = k \Rightarrow k = 27$	M1(AO 1.1) M1(AO 2.1) A1(AO 1.1) E1(AO 2.2a) M1(AO 3.1a) A1(AO 1.1) [6]	calculating det M setting their $2p - 26$ equal to 0 and solving conclude that $p = 13$ Deducing $R_3 = 3R_1 + 2R_2$ 27 from correct reasoning	OR M1 state that singular matrix requires each row to be a combination of the other two M1 state that $R_3 = 3R_1 + 2R_2$ or equivalent A1 conclude that $p = 13$	
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$\begin{pmatrix} 2a-3 & 2 \\ 2 & 5 \end{pmatrix}$	B1	I or 3I seen or used	
		i		B1	2 elements correct	
		i		B1	Other 2 elements correct Examiner's Comments This was answered correctly by the majority of candidates, the most frequent errors being not doubling one element of A or using I rather than 3I. Some candidates used $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ for I.	
		ii	$\frac{1}{4a-1}\begin{pmatrix} 4 & -1 \\ -1 & a \end{pmatrix}$ or equivalent	B1	Divide by correct determinant	
		ii		B1	Both diagonals correct Examiner's Comments The most frequent error was the omission of the determinant, while others made the leading diagonal negative instead of reversing the elements. Few candidates checked their answer by matrix multiplication of their answer with matrix A.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16				M1	Show correct expansion process for correct 3 × 3	
				M1	Correct evaluation of any 2 × 2	
			$3\lambda^2 - 7\lambda + 2$	A1	Obtain correct 3 term quadratic	
			$\frac{1}{3}$ or 2	B1*	Equate their det to 0	
				DM1	Attempt to solve a quadratic equation	
				A1	Obtain correct answers	
					Examiner's Comments The method of finding the determinant of a 3 × 3 matrix was well understood, with sign errors causing the majority of the loss of marks. Most knew that no unique solutions meant that they had to equate their determinant to zero, although some solved det M ≠ 0 or det M > 0.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	$\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$	B1 B1	Each column correct Examiner's Comments The correct matrix was found by most candidates.	
		ii	*Either* *Or*		*Either* *Or*	
		ii	P: $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$	B1 DB1	Stretch, s.f. 2 in y direction Shear, x-axis invariant e.g. $(0,1) \longrightarrow (2,1)$	
		ii		B1	Correct matrix	
		ii	Q: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$	B1 DB1	Shear, x axis invariant e.g. $(0, 1) \longrightarrow (1, 1)$ Stretch, s.f.2 in y direction,	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		B1	Correct matrix N.B. "in the x/y axis" is incorrect Examiner's Comments This part was found quite testing by many candidates. Some did not clearly indicate in their solution the identity of P and Q and the examiners had to assume that the first one given was P and the second one Q. The stretch was usually described correctly, but the shear was not. The expression "in the y-axis" would be part of the description of a reflection not a stretch. Candidates should be made aware that a shear has an invariant line, either the x- or y- axis, and that the image of one point must be stated correctly; "scale factor 2" is not acceptable when describing a shear. If transformation Q was given as the shear, its matrix was often $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ incorrect, i.e., and very few checked by matrix multiplication that their pair of matrices gave the answer to (i).	
		iii		M1	Attempt at matrix multiplication of two 2 × 2 matrices from (ii)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Obtain correct result cao Examiner's Comments Most candidates showed that they understood matrix multiplication, but many obtained the same answer as (i), thus not realising that this pair of transformations is not commutative.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	$(5a - 3b \ 10 \ 0)$	B1	2 elements correct, must be a 1×3 matrix (not coordinates)	
		i		B1	3rd element correct Examiner's Comments Most candidates found the correct answer, with $5a - 3b$ becoming $2a$ the most common error.	
		ii		M1	Single value	
		ii	$(6b - 5)$	A1	Correct answer, must be a matrix Examiner's Comments Many candidates thought this product gave a 3×1 or a 3×3 matrix. Those who correctly recognised the answer was a 1×1 matrix often omitted the matrix brackets.	
		iii	$\begin{pmatrix} 6a & 12 & 18 \\ 4a & 8 & 12 \\ -a & -2 & -3 \end{pmatrix}$	M1	Obtain a 3×3 matrix	
		iii		A1	All elements correct Examiner's Comments The correct 3×3 matrix was usually found.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	Shear,	B1	Must be a shear, otherwise 0/2.	
		i	x-axis invariant and e.g. (0, 1) → (3, 1)	B1	For invariant only allow parallel to or along x-axis, in x direction, for image allow 0.322°, 18.4°, $\tan^{-1}(1/3)$ or the complement, ignore scale factor if all OK otherwise. Column vectors for coordinates OK Examiner's Comments Most recognised that the transformation was a shear, but often did not clearly state that the x-axis is invariant or give a suitable image point under this transformation.	
		ii	*Either* $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -3 & -1 \\ -1 & 0 \end{pmatrix}$ or PQ = M	B1	Obtain correct matrix equation	
		ii		M1	Sensible attempt at multiplication of a pair of 2 × 2 matrices	
		ii	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	A2	Obtain correct answer, A1 for 3 elements correct	
		ii	reflection in $y = -x$	B2	Must be describing Q If they say M = QP → $P^{-1}M$ = Q, they can only get (possibly) M1 B2 if they obtain “correct” matrix for Q	
		ii	*Or* $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	B2	Diagram showing at least unit square and image under M coordinates shown	
		ii		B2	State reflection in $y = -x$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		B2	Correct matrix Examiner's Comments The most common error was using M = QP. The resulting matrix for P is then clearly not a single transformation, but candidates did not see that an error had probably occurred.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i		M1	Show correct expansion process for 3 × 3 condone sign errors	
		i	$a^2 - 8a + 15$	M1	Correct expansion for any 2 × 2, may use Cramer's rule which is M2	
		i		A1	Obtain given answer correctly Examiner's Comments The method for finding the determinant of a 3 × 3 matrix was demonstrated by virtually all candidates.	
		ii	det X = $3^2 - 3 \times 8 + 15 = 0$	B1	And must state not a unique solution or equivalent	
		ii		M1	Attempt to solve equations	
		ii	e.g. $3y + 5z = 5$, or $3x - 7z = -4$, or $5x + 7y = 5$	A1	Find a repeated equation and state consistency N.B. They may solve the equations first then deduce consistent and non-unique and gets 3/3 (possibly) Examiner's Comments Many candidates simply stated that det X = 0, without showing that this is true for the value a = 3. Most candidates attempted to solve the system of equations, but many made algebraic slips and so could not deduce the correct conclusion.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	$\frac{1}{2}\begin{pmatrix}1 & -a\\ 0 & 2\end{pmatrix}$ or equivalent	B1	Both diagonals correct	
		i		B1	Divide by correct determinant Examiner's Comments This question was answered correctly by most candidates. Omission of the determinant was the most common error.	
		ii	Either $P = BA^{-1}$	B1	State or use correct expression for P	
		ii	$\begin{pmatrix}1 & 0\\ 2 & 1-2a\end{pmatrix}$	M1	Multiplication attempt, 2 elements correct for any pair of matrices	
		ii		A1ft	Obtain correct answer a.e.f. ft for their (i)	
		ii	Or Using PA = B	B1	State or find correct 1st column of P	
		ii		M1	Multiplication attempt to find "$1 - 2a$"	
		ii		A1	Obtain completely correct answer Examiner's Comments This question was generally answered correctly. The most common error was finding $A^{-1}B$.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i		B1	Coordinates of any 2 images seen	Might be columns
		i	A' (0, −1) B' (2,− 1) C' (2, 0)	B1	Coordinates of 3rd image seen	
		i		B1	Completely correct labelled diagram, must include indication of coordinates Examiner's Comments Many candidates did not show sufficient detail. Coordinates were not clearly displayed or scales shown clearly on the axes to indicate them. Some just plotted the image points, thus not showing that the image of the unit square is a rectangle.	
		ii	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$	B1	Rotation and stretch or vice versa	
		ii		B1	Rotation 90° clockwise, then Stretch s.f. 2 parallel to x-axis Or Stretch s.f. 2 parallel to y-axis & Rotation 90°	Must be a correct pair in correct order Consistent with their pair of transformations
		ii		B1ft	clockwise	
		ii		B1ft	Correct matrix, Correct matrix	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Or $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$		S.C. If 1 matrix correct, correct 2nd matrix can be found by matrix multiplication and not be necessarily consistent with their transformation, but not ft. Examiner's Comments Most candidates identified that the pair of transformations was a rotation and a stretch, but made errors in the direction of rotation or stretch. Many who got a correct pair of transformations could then not give the correct matrix that represented them.	Just a trig form for rotation not acceptable
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i		M1	Attempt to find det D	
		i		M1	Show correct process for a 3 × 3, condone sign errors	Or Cramer's rule or similar
		i		M1	Show correct processes for a 2 × 2	
		i	$a^2 - 6a + 5$	A1	Obtain correct answer	
		i		M1	Attempt to solve det D = 0	
		i	a = 5 or 1	A1	Obtain correct answers Examiner's Comments Most candidates realised that the determinant had to be found and apart from minor sign errors were able to find, correctly, the values for singularity. However, those who solved a quadratic equation and found non-integer answers generally did not go back to see if an error had been made somewhere.	
		ii		B1	State unique solution	
		ii		B1	State non unique solutions	
		ii		M1	Attempt to solve equations with a = 1	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	Explain inconsistency with correct working S.C. Answer to (i) wrong, allow correct unique / non-unique B1ft, B1ft only Examiner's Comments This part prove quite testing. The value of the determinant needs to be considered first as this determines uniqueness or not. If non-unique then the solution of the equations must be investigated to determine if they are consistent or not.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	(7 23)	B1B1	Each element correct, missing brackets B1 only Examiner s Comments A row matrix was found by the majority of candidates, with only simple arithmetic errors occurring in a few cases. A few gave the answer as a column matrix, while others obtained a 1 × 1 matrix such as (30).	
		ii	$\begin{pmatrix} 6 & -15 \\ 4 & -10 \end{pmatrix}$	M1	Obtain 2 × 2 matrix	
		ii		A1	Obtain 2 correct elements	
		ii		A1	Obtain other 2 correct elements	
		ii	det CB = 0	A1FT	Obtain their det CB, must be a 2 × 2 matrix	
		ii	singular	A1FT	Correct conclusion from their det CB Examiner s Comments A significant minority found BC or thought that CB was a 1 × 1 matrix. Of those who found a 2 × 2 matrix, only a few made odd arithmetic slips and most candidates found their determinant correctly and deduced singularity or non-singularity correctly. The most common error was to state non-singularity having found the determinant to be zero.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
25				B1	Establish result true for n = 1 or n = 2	
				M1	Multiply M and M^k, either order	
			$2\left(2^{k+1}-2\right)+2$ or $2^{k+1}+2^{k+1}-2$	A1	Obtain correct element	
				A1	Obtain other 3 correct elements	
				A1	Obtain $2^{k+2}-2$ convincingly	
				B1	Specific statement of induction conclusion, provided 5/5 earned so far and verified for n = 1 Examiner's Comments This was one of the less well-attempted questions. A significant minority thought that the induction step required the addition of M^k and M. Many failed to show sufficient working either in establishing the validity when n = 1, or in obtaining the elements in M^{k+1}. Centres should remind candidates of this, as well as the need for a clear statement of the induction conclusion.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	B1B1	Each column correct Examiner's Comments The majority of candidates obtained the correct matrix. Those with a wrong matrix usually failed to show any working; a sketch of the unit square under the required transformation might have helped.	
		ii	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	B1B1	Each column correct Examiner's Comments This part was done slightly better than part (i), but the above comment also applies to this part.	
		iii	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	M1	Attempt at matrix multiplication in correct order	
		iii		A1FT	Obtain correct answer from their (i) and (ii) Examiner's Comments The most common error here was to multiply the matrices in the wrong order.	
		iv	Reflection, in $y = x$	B1B1	Correct description of their (iii) only Examiner's Comments Most candidates were able to describe correctly the transformation represented by their answer to part (iii).	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i		M1	Show correct expansion process for 3 × 3	
		i		M1	Correct evaluation of any 2 × 2	
		i	$a + 3$	A1	Obtain correct answer	
		i		M1	Use det A = 0	
		i	$a = -3$	A1FT	Obtain correct answer from their det A Examiner's Comments The determinant was found and equated to zero by most candidates, with only minor arithmetic slips being seen.	
		ii	$\frac{1}{a+3}\begin{pmatrix} 1 & -1 & 1 \\ 7 & a-4 & 1-2a \\ -11 & 8-a & 3a-2 \end{pmatrix}$ $\frac{1}{a+3}\begin{pmatrix} 2 \\ 2-4a \\ 7a-1 \end{pmatrix}$	M1 A1 A1 B1 M1 A2	Show correct processes for adjoint entries Obtain at least 4 correct entries in adjoint Obtain completely correct adjoint Divide adjoint by their det A Pre-multiply column matrix by their A^{-1} Obtain correct answer, A1 for 1 element correct	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			Examiner's Comments Although most candidates knew the various stages that are required when finding the inverse of a 3 × 3 matrix, often each stage of the working was labelled as A^{-1}. This was not penalised, but centres should try to get candidates to annotate their work in a more precise way. Having found an inverse matrix, a number of candidates then did not attempt to use the inverse to solve the equations and others incorrectly post-multiplied by the inverse. Many made simple algebraic or arithmetic slips at this stage, or omitted the determinant completely.	
			Total	12		

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance	
28				M1	Show correct expansion process for 3 × 3	Condone sign errors for first M1 M2 for the "diagonal" method
				M1	Correct evaluation of any 2 × 2	
			$2a^2 + 6a - 15$	A1	Obtain correct answer i.s.w. <u>Examiner's Comments</u> Almost all candidates showed a correct expansion method, with only minor arithmetic or algebraic errors occurring. Some found the correct value of the determinant, but then gave its reciprocal, as if they were finding the inverse matrix, which a few candidates actually did thus making the solution much longer.	Det = $1/(2a^2 + 6a - 15)$ only A0
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i		B1	2 elements correct	Condone missing brackets in (i) (ii) & (iii)
		i	$\begin{pmatrix} 7 & 3 \\ -18 & 19 \end{pmatrix}$	B1	All elements correct Examiner's Comments This was generally answered correctly. The two most common errors were finding B + 2I and subtracting this from 4A,	
		ii		B1	while some thought I was Both diagonals correct, ignore determinant Examiner's Comments Omission of the determinant was the most common reason for loss of marks.	
		ii	$\frac{1}{14}\begin{pmatrix} 5 & -1 \\ 4 & 2 \end{pmatrix}$ or equivalent	B1	Divide their matrix by correct determinant	$\frac{\begin{pmatrix} 5 & -1 \\ 4 & 2 \end{pmatrix}}{14}$ is OK for 2[nd] B1
		iii	$(AB^{-1})^{-1} = BA^{-1}$ or $\mathbf{B}^{-1} = \frac{1}{7}\begin{pmatrix} 3 & -1 \\ -2 & 3 \end{pmatrix}$	B1	Correct result seen or used	
		iii		M1	Multiplication attempt for any pair of 2 × 2 matrices, 2 elements correct, but not I	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\frac{1}{14}\begin{pmatrix}19 & -1\\ 22 & 4\end{pmatrix}$	A1	Correct answer a.e.f. Examiner's Comments Many candidates did not know the result for the inverse of a product of two matrices and found the value of $A^{-1}B$. A significant number of candidates found B^{-1}, then AB^{-1} and then attempted the inverse of this, usually making an arithmetic error somewhere or not being able to find the determinant of AB^{-1} correctly.	
			Total	7		
30		i		B1B1	Rotation, 45° or $\pi/4$ clockwise or equivalent Examiner's Comments The transformation being a rotation was generally recognised, the difficulty was in giving a consistent direction and angle pair.	Must be rotation and no other transformation, otherwise 0/2
		ii	(det X =) 1	B1	Correct value	
		ii		B1ft	Scale factor for area or equivalent Examiner's Comments The correct value was generally found and most gave a satisfactory explanation of its connection to the area of a transformed shape.	e.g. area unchanged
			Total	4		